

Test Evidence for Greenfield Local Hub (GLH)

Version 1 Tests

Planned development:

- 1.1 Back-end Database
- 1.2 User Registration
- 1.3 User Login
- 1.4 User Logout

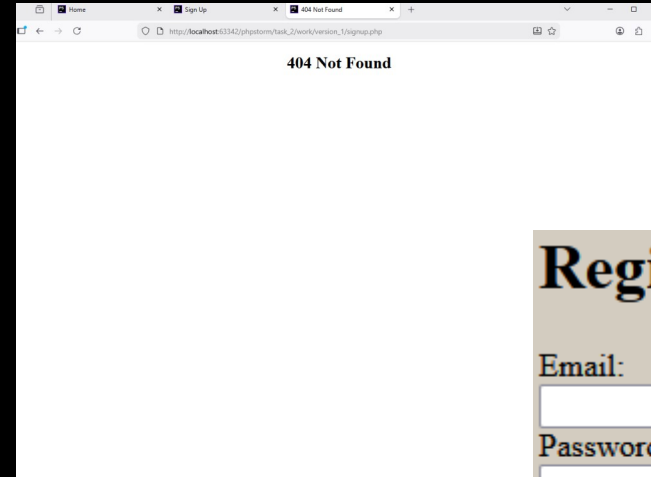
Test no. 1.1 Register Test

Test data:

email: emma.wilson@testmail.local

password: TestPass!2024

Tests resulted in a 404 error, this could mean Multiple issues have occurred. After fixing file Path names. Test was conducted again resulting in an error displaying “ERROR: could not find driver”. This indicates an issue with The PDO connection to the database.



Register

Email:

Password:

Confirm Password:

Sign Up

ERROR: could not find driver

Test no. 1.1 Register Test

During this test, a technical error was found. Laragon (database) had some errors which had to be resolved by my tutor (the issue was technical, allowing assistance). There was a missing file in the laragon version (php.ini) which prevented the code from running. After the fix the code ran as intended. Redirecting the user to the login page.

```
print_r(PDO::getAvailableDrivers());
```

Array ()

My tutor had to submit to my registration page to ensure that the system did not come back with the error again. Due to my tutor not knowing my planned test data, he used his own as follows below:

Email: test@test.com

Password: test

The system worked as intended with the password being hashed for security and the system including a sign-up date.

	user_id	email	password	sign_up_date
1		test@test.com	\$2y\$10\$K3/LQsgWKwSPpns74UtzOotWZiPU4E7OyzJS3j1PfZ...	2026-03-23

Test no. 1.1 Register Test

After the technical issue was resolved, I did the intended test again with the correct test data (see page 3).

On submit, the system redirected the user to the login page to login to the system with a message displayed to the user saying "ACCOUNT CREATED SUCCESSFULLY". This tells the user the account has been created. After checking the database, the test was displayed there as well with the password hashed and with the correct sign up date as intended.

USER CREATED SUCCESSFULLY

Login

Email:

Password:

		user_id	email	password	sign_up_date
☐	Edit Copy Delete	1	test@test.com	\$2y\$10\$K3/LQsgWKwSPpns74UtzOotWZiPU4E7OyzJS3j1PfZ...	2026-03-23
☐	Edit Copy Delete	2	emma.wilson@testmail.local	\$2y\$10\$6odMerJOf91WzApXWBKcRuQn5.hNo1V/JRhv4.PlsmX...	2026-03-23

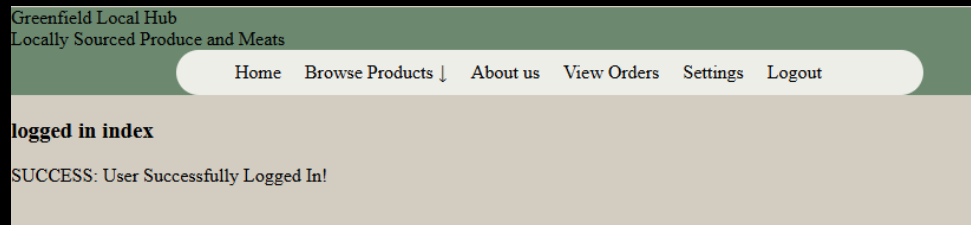
Test no. 1.2 Logout Test

Test data:

email: emma.wilson@testmail.local

password: TestPass!2024

Test resulted in user successfully being logged out and sent back to the index page with a message displaying the action. The navigation bar was also successfully updated to not being logged in as intended.



Version 2 Tests

Planned development:

- 2.1 Audit System
- 2.2 Product Display
- 2.3 Basket
- 2.4 Checkout
- 2.5 View Orders

Test no. 2.1 Basket Test

Your Basket

Below are the Items in your basket

Name	Price	Sub-Total	quantity	Actions
Homemade Beef Pie (400g)	£5	£ 10	2	Update Quantity Delete
Fresh Sourdough Bread (500g)	£2.8	£ 2.8	1	Update Quantity Delete
Double Cream (300ml)	£1.9	£ 7.6	4	Update Quantity Delete
Artisan Cheddar Cheese (200g)	£3.5	£ 3.5	1	Update Quantity Delete

Your Basket Total

Your Basket Subtotal: £23.9 Select delivery or collection date:

After an item is added in the basket, it is successfully added to the basket, where a user can change the quantity or delete an item.

After a couple tries and some editing, this error message appeared on basket.php. To fix it, all that had to be done was create a basket if one didn't already exist. The system had an issue where if you tried to click on the basket before adding an item, it wouldn't work. This issue has now been resolved.

Warning: Undefined array key "basket" in H:\phpstorm\task_2\work\version_2\assets\common.php on line 110

Fatal error: Uncaught TypeError: array_key_exists(): Argument #2 (\$array) must be of type array, null given in H:\phpstorm\task_2\work\version_2\assets\common.php:110 Stack trace: #0 H:\phpstorm\task_2\work\version_2\products.php(21): in_basket('1') #1 {main} thrown in H:\phpstorm\task_2\work\version_2\assets\common.php on line 110

Test no. 2.2 Order Test

Basket

Below are the Items in your basket

Name	Price	Sub-Total	quantity	Actions
Artisan Cheddar Cheese (200g)	£3.5	£ 3.5	1	Update Quantity Delete
Homemade Beef Pie (400g)	£5	£ 5	1	Update Quantity Delete

Your Basket Total

When testing I ran into an issue, When a user would place an order, the order id and product ids would flip. The product id would be where the order id should be and the order id would be where the product id should be.

This is a serious problem as it assigns products to incorrect orders and changes what product a user has ordered.

After investigating the issue, I discovered that it was caused by the orderid and itemid (productid) being flipped whilst executing the insert statement. (see page 10)

Orders table:

☐	Edit	Copy	Delete	5	1 col	2026-04-01	2026-03-30
--------------------------	------	------	--------	---	-------	------------	------------

Basket table:

☐	Edit	Copy	Delete	6	5	2	1	2026-03-30
☐	Edit	Copy	Delete	7	5	3	1	2026-03-30

Test no. 2.2 Order Test

```
function add_basket($conn, $basket, $orderid){  
    $productstmt = $conn->prepare("INSERT INTO basket (product_id, order_id, quantity, date) VALUES (?, ?, ?, NOW())");  
    //prepares an sql statement. NOW() gets the current date to insert  
  
    foreach ($basket as $itemid => $quantity){  
        $productstmt->execute([$orderid, $itemid, $quantity]);  
    }  
}
```

After the order and item id were flipped with the item id first, the issue was resolved and the data was implemented as intended after retesting.

Orders table:

☐	Edit	Copy	Delete	6	1 col	2026-04-01	2026-03-30
--------------------------	------	------	--------	---	-------	------------	------------

Basket table:

☐	Edit	Copy	Delete	8	2	6	1	2026-03-30
☐	Edit	Copy	Delete	9	3	6	1	2026-03-30

Version 3 Tests

Planned development:

- 3.1 View, Alter and Delete Orders
- 3.2 Loyalty Scheme
- 3.3 Stock Management
- 3.4 System testing

Test no. 3.1 Remove stock Test

				basket_id	product_id	order_id	quantity	date
☐	Edit	Copy	Delete	1	2	1	1	2026-03-27
☐	Edit	Copy	Delete	2	2	2	3	2026-03-27
☐	Edit	Copy	Delete	4	3	1	1	2026-03-30
☐	Edit	Copy	Delete	5	4	2	1	2026-03-30
☐	Edit	Copy	Delete	6	5	2	1	2026-03-30
☐	Edit	Copy	Delete	7	5	3	1	2026-03-30
☐	Edit	Copy	Delete	8	2	6	1	2026-03-30
☐	Edit	Copy	Delete	9	3	6	1	2026-03-30
☐	Edit	Copy	Delete	10	1	7	-1	2026-03-30
☐	Edit	Copy	Delete	11	2	7	1	2026-03-30
☐	Edit	Copy	Delete	12	1	8	-1	2026-03-30
☐	Edit	Copy	Delete	13	2	8	1	2026-03-30
☐	Edit	Copy	Delete	14	3	8	3	2026-03-30
☐	Edit	Copy	Delete	15	4	8	4	2026-03-30
☐	Edit	Copy	Delete	16	1	9	-1	2026-03-30
☐	Edit	Copy	Delete	17	2	9	1	2026-03-30
☐	Edit	Copy	Delete	18	3	9	3	2026-03-30
☐	Edit	Copy	Delete	19	4	9	4	2026-03-30
☐	Edit	Copy	Delete	20	1	10	-1	2026-03-30
☐	Edit	Copy	Delete	21	2	10	3	2026-03-30
☐	Edit	Copy	Delete	22	3	10	1	2026-03-30

```
function remove_stock($conn, $post){  
    $sql = "UPDATE products SET quantity = quantity - ? WHERE product_id = ?";  
    $stmt = $conn->prepare($sql);  
    $stmt->bindparam(1, $post['quantity']);  
    $stmt->bindparam(2, $post['product_id']);  
  
    $stmt->execute();  
    $conn = null;  
    return true;  
}
```

```
function remove_stock($conn, $post){  
    $sql = "UPDATE basket SET quantity = quantity - ? WHERE product_id = ?";  
    $stmt = $conn->prepare($sql);  
    $stmt->bindparam(1, $post['quantity']);  
    $stmt->bindparam(2, $post['product_id']);  
  
    $stmt->execute();  
    $conn = null;  
    return true;  
}
```

This proves the importance of test data and testing the system before you release it for public use. As if this issue occurred in a live system, a large amount of customer order data would have been destroyed

Whilst I was testing my remove_stock function, I had accidentally set the table to basket instead of products. This resulted in the quantities of all products in the basket with the product_id of 1 to have their quantities altered.

Test no. 3.1 Remove stock Test

```
1 usage new *
function remove_stock($conn, $basket){
    $stockstmt = $conn->prepare("UPDATE products SET quantity = quantity - ? WHERE product_id = ?");

    foreach ($basket as $itemid => $quantity){
        $stockstmt->execute([$quantity, $itemid]);
    }
    $conn = null;
    return true;
}
```

In the end, I had formatted the function incorrectly. It needed to be as shown above, not like how it is on page 12. The function now goes through each item that has been ordered and removes the quantity from the quantity available.

Test no. 3.2 Loyalty Scheme Test

	loyalty_id	user_id	points
☐ Edit Copy Delete	1	6	10

	loyalty_id	user_id	points
☐ Edit Copy Delete	1	6	15

Your Basket Total

Your Basket Subtotal: £5.98

Points Earned: 5

Please select an option: Collection ▾

Select delivery or collection date:

15 / 04 / 2026 📅

Delete Basket

Complete Order

You have 15 loyalty points

For Adding points on purchase. The system stores points in session based on the basket subtotal so 1 point every £1 spent which would be redeemed at checkout. Add points function worked as intended. Updating a users points when they make a purchase.

Test no. 3.3 Cancelling Orders Test

Your Dashboard

You have **39** loyalty points

SUCCESS: Your Order has been completed

Your Dashboard

You have **29** loyalty points

Warning: foreach() argument must be of type array/object, string given in H:\phpstorm\task_2\work\version_3\assets\common.php on line 238

Greenfield Local Hub
Locally Sourced Produce and Meats

Home Browse Products ↓ About us View Orders Settings Logout

SUCCESS: Appointment has been cancelled.

Your Orders

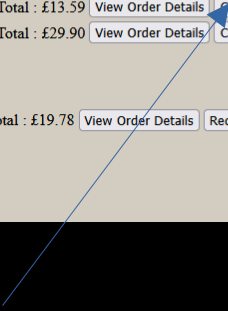
Below are your orders:

Current orders

Delivery Date: 2026-04-04 Ordered on: 2026-03-31 For : Delivery Total : £13.59	View Order Details	Cancel Order	Edit Order
Delivery Date: 2026-04-01 Ordered on: 2026-03-31 For : Collection Total : £29.90	View Order Details	Cancel Order	Edit Order

Past orders

Delivery Date: 2026-03-27 Ordered on: 2026-03-31 For : Delivery Total : £19.78	View Order Details	Request Refund
--	--------------------	----------------



The loyalty points were updated but the order wasn't actually cancelled. See 16 for solution

Test no. 3.3 Cancelling Orders Test

Warning: Array to string conversion in H:\phpstorm\task_2\work\version_3\orders.php on line 20

Warning: Array to string conversion in H:\phpstorm\task_2\work\version_3\orders.php on line 20

Warning: Array to string conversion in H:\phpstorm\task_2\work\version_3\orders.php on line 20

Warning: Array to string conversion in H:\phpstorm\task_2\work\version_3\orders.php on line 20

Warning: foreach() argument must be of type array/object, string given in H:\phpstorm\task_2\work\version_3\assets\common.php on line 238
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Locally Sourced Produce and Meats

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SUCCESS: Appointment has been cancelled.

Your Orders

Below are your orders:

Current orders

Delivery Date: 2026-04-01 Ordered on: 2026-03-31 For : Collection Total : £29.90 [View Order Details](#) [Cancel Order](#) [Edit Order](#)

Past orders

Delivery Date: 2026-03-27 Ordered on: 2026-03-31 For : Delivery Total : £19.78 [View Order Details](#) [Request Refund](#)

To fix the issue and allow orders to be cancelled, I had to change from getting the order_id through session (causing truancy in the array) to getting them through post.

```
try {
    //put stock check and remove loyalty points here
    $points = implode(display_loyalty(dbconn_select(), $_SESSION['user'])) - $_SESSION['totalpoints']; //gets the points to remove

    if (remove_points(dbconn_update(), $_SESSION['user'], $points)) { //removes the points, if it comes back true then moves on

        $stock = implode(find_stock(dbconn_select(), $_SESSION['order_id'])); //gets the stock from all the items in the order

        if (add_stock(dbconn_update(), $stock)) { //adds the stock back to be available, if it comes back true allows order to be deleted

            if (cancel_ord(dbconn_delete(), $_SESSION['order_id'])) { //deletes the order from the database

                $_SESSION['usermessage'] = "SUCCESS: Appointment has been cancelled."; //success message displayed to the user on a successful cancel
                auditor(dbconn_insert(), $_SESSION['user'], "ord", "User has successfully cancelled order."); //audit to the system on order cancell
            } else {
                $_SESSION['usermessage'] = "ERROR: Could not be able to execute complete this action"; //displays failure message to the user
            }
        }
    }
}
```

```
if (isset($_POST['orddelete'])) {
    try {
        //put stock check and remove loyalty points here
        $points = implode(display_loyalty(dbconn_select(), $_SESSION['user'])) - $_SESSION['totalpoints']; //gets the points to remove

        if (remove_points(dbconn_update(), $_SESSION['user'], $points)) { //removes the points, if it comes back true then moves on

            $stock = implode(find_stock(dbconn_select(), $_POST['order_id'])); //gets the stock from all the items in the order

            if (add_stock(dbconn_update(), $stock)) { //adds the stock back to be available, if it comes back true allows order to be deleted

                if (cancel_ord(dbconn_delete(), $_POST['order_id'])) { //deletes the order from the database

                    $_SESSION['usermessage'] = "SUCCESS: Appointment has been cancelled."; //success message displayed to the user on a successful cancella
                    auditor(dbconn_insert(), $_SESSION['user'], "ord", "User has successfully cancelled order."); //audit to the system on order cancellat
                } else {
                    $_SESSION['usermessage'] = "ERROR: Could not be able to execute complete this action"; //displays failure message to the user
                }
            }
        }
    }
}
```


Version 4 Tests

Planned development:

- 4.1 Staff Login System
- 4.2 Registering Staff Accounts
- 4.3 Updating Product Quantities
- 4.4 Adding New Products

Test no. 4.2.1 Registering to the system

To start my testing, I need to start on the register system so that an account can be created to test the system login.

The initial test showed the error below:

```
ERROR: SQLSTATE[HY093]: Invalid parameter number: number of bound variables does not match number of tokens
```

This means there is likely an issue with the values in my sql statement (forgot to update when pulling from the customer side)



```
function reg_user($conn, $post){  
    $sql = "INSERT INTO staff (email, type, password, sign_up_date) VALUES (?, ?, ?)"; //sql statement
```

As shown above, There is only 3 ? in values when there needs to be 4.

This will be updated, as well as checking other statements to ensure the values are all up to date.

Test no. 4.2.2 Registering to the system

After the parameter issue was resolved, I retested and got another error message:

```
ERROR: SQLSTATE[23000]: Integrity constraint violation: 1048 Column 'staff_id' cannot be null
```

Originally, I wasn't quite sure what was causing this error but after checking my database. I discovered that an account has successfully been made. This means the error is likely being caused by the audit system.

		▼ staff_id	email	type	password	sign_up_date
☐	 Edit  Copy  Delete	1	test@greenfield.com	adm	\$2y\$10\$Ht/6p6SFMMrkWK7qJXrsk.NXvND05tY0Y40uW3DI.d8...	2026-04-01

Instead of fixing this issue now, I will be fixing it after the login system works as only admins will be able to register new accounts so the audit should have their accounts id not the registered account. I will be moving onto the login system since a staff account now exists with a properly hashed password.

Test no. 4.1.3 Login system

During the initial test of the login system, I got the same error I received with the register system.

```
ERROR: SQLSTATE[23000]: Integrity constraint violation: 1048 Column 'staff_id' cannot be null
```

Login

This proves that the issue with the audit system is more severe then before and needs looking at now rather then later.

```
$_SESSION['user'] = $usr['user_id']; //checks if user matches the user id
```

Whilst investigating, I discovered this. I changed 'user_id' to 'staff_id' and it worked as intended with an audit being logged to the system.

	audit_id	staff_id	code	long_desc	date
☐ Edit <input type="plus"/> Copy <input type="minus"/> Delete	1	1	log	User has successfully logged in	2026-04-01

Test no. 4.2.4 Register system

Thankfully During the test on the register system after fixing login, It worked as intended with an account being successfully made with the audit recording the admin account that made it (staff_id = 1) and not the account that was just made

USER CREATED SUCCESSFULLY

		staff_id	email	type	password	sign_up_date
☐	Edit Copy Delete	1	test@greenfield.com	adm	\$2y\$10\$Ht/6p6SFMMrkWK7qJXrsk.NXvNDo5tY0Y40uW3DI.d8...	2026-04-01
☐	Edit Copy Delete	2	store@greenfield.com	sto	\$2y\$10\$vQY5Xz8lQkOa83tX5QgVD.PfgsPjTywomy/fp3qrzNA...	2026-04-01

		audit_id	staff_id	code	long_desc	date
☐	Edit Copy Delete	1	1	log	User has successfully logged in	2026-04-01
☐	Edit Copy Delete	2	1	reg	New staff account created	2026-04-01

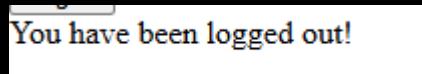
Test no. 4.5.5 Logout system

A system I failed to plan for was a users ability to log out of their account. Another oversight on my part. I was originally going to use the same login that the regular users use to reduce code repeatability but that was not the most secure option as I want to keep the staff and user common and system completely separate so I chose to repeat the code to accomplish that and to ensure the audits were sent to the correct table and users were sent to the correct pages.

On logout, The audit was made successfully but no message was displayed to the user confirming the logout.



After investigating, I discovered that the login page did not have the correct message display code for when users aren't logged in so it didn't display. It has now been corrected.



Test no. 4.4.6 Adding new products

Whilst testing my get farms function, to display the farms stored in the database the farms weren't displaying and I got the following error

```
ERROR: SQLSTATE[42S02]: Base table or view not found: 1146 Table 'glh_db.farm' doesn't exist
```

This was caused by forgetting an s in the sql statement.

```
usage new *  
unction get_farms($conn){  
    $sql = "SELECT * FROM farm WHERE 1"; //sql statement  
    $stmt = $conn->prepare($sql); //prepares to sql  
    $stmt->execute(); //executes statement  
    $result = $stmt->fetchall(PDO::FETCH_ASSOC); //returns the result as an associative array  
    $conn = null; //closes connection so it cant be abused  
    return $result; //returns the result
```

This has now been fixed.

Test no. 4.4.6 Adding new products

During my first test, an error message showed an issue with the image alt tag. It wasn't sending to the database. I checked to make sure the naming conventions were right and they weren't after this was fixed the code ran as expected with data being added to the database successfully.

```
ERROR: SQLSTATE[23000]: Integrity constraint violation: 1048 Column 'image_alt' cannot be null
```

```
<label for='imgalt'>Image Alt Tag</label><br>"; //label for  
<input type='text' name='imgalt' id='imgalt' required><br>";
```

```
$stmt->bindParam(5, $post['img_alt']);
```

☐	Edit	Copy	Delete	15	1 pro	User has added a new product	2026-04-02
--------------------------	------	------	--------	----	-------	------------------------------	------------

☐	Edit	Copy	Delete	11 fresh Baguette	[INSERT DESCRIPTION HERE] GVP	assets/images/baguette.png	a fresh baguette	2.3	25
--------------------------	------	------	--------	-------------------	-------------------------------	----------------------------	------------------	-----	----

One thing I did not think about was actually getting the image stored in the correct file so as future development, I would develop a drag and drop to drop the image file in and send to the assets folder.